Cover Letter

From: **Matthew Mohebbi, CEO**
Information Technologies Curves, Inc. dba IT Curves
8201 Snouffer School Road
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Email: mmohebbi@itcurves.net, Phone: (301) 793-2440

Date: September 06, 2024

To: **Grant Transit Authority**
Office of Procurement
Attn: Eric Loomis
Email: gta@granttransit.com

Subject: Response to RFP #24-01 TRANSIT ROUTE SCHEDULING/RUNCUTTING SOFTWARE SYSTEM

Dear Mr. Loomis,

It is our pleasure here at IT Curves to provide you with our response to RFP #24-01, Transit Route Scheduling/Runcutting Software System. This proposal describes the systematic and strategic approach to address challenges faced by any transit agency in planning and executing fixed route transportation services.

- IT Curves software solution will not only provide enough tools to intelligently and conveniently manage fixed route planning and optimization, in addition will bring in additive advantage of integrating with currently deployed components which are already being utilized by GTA for scheduling and dispatch.
- IT Curves planning software will be able to utilize stop usage by pulling in the passenger counts coming from already installed and integrated APC equipment in GTA CAD AVL.

The prime contractor and the system integrator – IT Curves, a Maryland corporation located in Montgomery County – is a leader in transit and transportation CAD and AVL, and systems integration. We have recently adapted and Integrated Artificial Intelligence and Machine Learning techniques into our Fixed Route Planning and CAD AVL system for route enhancements by studying the drive patterns and ridership on popular stops. Another recent research and advancement IT Curves system has done on the productivity of fixed route resulting in adaptation of flex/deviated route and micro transit solution. The recommendations are made based on the consensus data to identify population density areas as well as the passenger counts to identify the most utilized stops during different hours of the day. IT Curves has already utilized these enhanced modules and joint venture with City of Annapolis, piloted a micro transit service by replacing two fixed route buses.

GTA is currently utilizing IT Curves' suite of integrated software which includes CAD/AVL, route scheduling and dispatching. This in fact brings in additive advantage as proposed planning and cost optimization tools will be directly integrated with the current system and will be able to utilize the components and operational data to assist future route plannings. Having reviewed RFP, we can confidently say that our proposed system's enhancements and new system capabilities not only match or exceed all the requirements listed in your RFP but will help GTA reach the ambitious objectives of its core planning and route cost optimization.

This RFP opportunity is perfect for IT Curves as it is very similar in nature to others that we have both won and found a great deal of success with. As described above IT Curves has already been working in a similar direction and has already implemented part of the system to achieved similar systems objectives in recent years.

Treating our clients as partners is a major factor in how we operate. IT Curves will work closely with you to make sure that some proposed components are tailored to the GTA staff requirements and taste to make whole adaptation of new tools easy and smooth, and our highly customizable system is configured to serve your specific needs. Along with the new contract and new planning module, IT Curves is also committed to enhancement existing tools to make scheduling easier. IT Curves provides ongoing support for all our clients, including patches to resolve issues or to release additional tools and capabilities. Finally, the same IT Curves engineers who install your system and train your staff will lead your customer support team to ensure a continuity of service.

Feel free to contact Kevin Rivas for the purposes of this RFP. Kevin is leading IT Curves response and relationship with GTA. He will manage support throughout the life of the contract if awarded. His contact is as follows, krivas@itcurves.net, (301) 208-2222x3256.

We are hopeful that you will allow IT Curves to present, preferably in person, our solution prior to final selection to assist in your decision-making process. Our live presentation of our products and services, a full demonstration of their features, will give the GTA a strong understanding of who we are and how we operate.

IT Curves has included all forms necessary to submit a response to this RFP. We are fully prepared to submit any extra necessary documents for the successful bidder upon confirmation from Grant Transit Authority. Please do not hesitate to contact us should you have any questions or need any additional information after reviewing our proposal. We hope to earn your business and to be the technology provider and supporter for your transit service now and into the future!

Sincerely,

Matthew Mohebbi
CEO, IT Curves

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IT Curves FEIN: 26-2098733
IT Curves DUN: 063920114

This bid is subject to the negotiation of a contract on mutually agreeable terms following award. For any matters regarding the information in this proposal and its contents, please contact both the authorizing official and the secondary contact

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ITCURVES

INTELLIGENT TRANSPORTATION SOFTWARE

From Information Technologies Curves
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(301) 208-2222

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General Statement of Contractor Experience

IT Curves is pleased to present a comprehensive response to Grant Transit Authority's request for proposals for a Transit Route Scheduling/Runcutting Software System. Our proposal, and our Firm Fixed Price (FFP) for scope described in this proposal is valid for 180 days. Our response to RFP is keenly focused on addressing the unique fixed route planning challenges that includes scheduling and Runcutting. The proposed system will enable GTA planning staff to visually modify routes with additional proposed stops and convert fixed stops to OnDemand stops. The planning tool will utilize various metrices, such as increasing ridership on certain stops, consensus data at tract and block level to identify high population density areas. The proposed software will enable GTA planning staff to create different variations of the routes and compare the cost of route in terms of operating time and miles. The proposed software will not only bring in the additional requested feature for planning, Runcutting and rostering, in addition IT Curves will enhance existing software interfaces as per required by GTA team. IT Curves will do existing software enhancements at no cost. Both the newly proposed software and existing scheduling and dispatch software will be seamlessly integrated to avoid any duplicated effort on the operation side to organize new routes and make them available to the dispatch system. Within this proposal you can find a direct response to each criterion, IT Curves' understanding of the challenges at hand, and background on both our company and the project team. The proposal highlights our comprehensive support system that sets us apart from the others. offering more prompt and adaptable assistance compared to other offerors, ensuring that client needs are met swiftly and effectively. In this executive summary, we briefly highlight the most relevant subsystems and our approach to addressing some of the transit challenges faced in rural counties in your service area, and our system solution to those.

- Tools to visualize planned routes and runs for driven miles and times
- Compare the deadhead vs available to rider's miles among different models
- Identify and remove fix schedule for underutilized stops
- Adjust route plan by moving, adding and removing stops without impacting current service
- View and compare different variation in route before publishing them
- Runcutting for future use
- Rostering for future use

1. Contractor Identification

IT Curves

8201 Snouffer School Rd, Gaithersburg, MD 20879
IT Curves FEIN: 26-2098733
IT Curves DUN: 063920114
Contact: Kevin Rivas (301) 208-2222, ext. 3256. krivas@itcurves.net

Located in Gaithersburg, MD, IT Curves is a premier provider of innovative and pioneering transportation technology solutions. As active members of the American Public Transit Association, The Transportation Alliance, and the Transportation Research Board, we are constantly developing new features and integrating the latest technologies into our off-the-shelf package of transportation management software.

IT Curves is a technology leader in Intelligent Transportation Systems (ITS), transportation automation, fixed bus planning and scheduling tools for applications such as:

- Advanced fixed-route scheduling and management module
- Advanced, in-vehicle AVA system
- MDTs for receiving routes, navigation, and safe communication with dispatch
- Continuous automatic monitoring to provide accurate AVL and forecasting for all trips, allowing for advanced intervention in case of delays
- On-demand trip capabilities allow trips to be added by dispatch to routes in progress, where capacity and time allows
- Microtransit system to enhance fixed routes with flexible curb-to-curb services to increase operation efficiency and help in first-mile, last mile solutions
- Commingling of paratransit services with microtransit service allows for usage of additional paratransit capacity, increasing service efficiency

Extensive customer support, not only to transit agencies, but in some cases to riders of transit agencies in usage of the rider app and portal as may be requested by an agency.

Kevin Rivas is the account manager assigned to this project, is a subject matter expert on fixed route planning, scheduling, runcutting, rostering and dispatching. Mr. Rivas was instrumental in the development and implementation of the simulation and planning tools utilized in Annapolis, MD. With Mr. Rivas' help, the city of Annapolis Transit has more effectively and efficiently planned routes, with metrics to display recent improvements to the transit network.

2. *Description of Qualifications (Staff)*

Matthew Mohebbi, CEO SPOTLIGHT

mmohebbi@itcurves.net, (301) 793-2440

Matthew Mohebbi is IT Curves' founder and CEO. Since learning of this procurement, he has taken great interest in GTA. He plans to participate in early discussions to ensure the project is properly kicked off in the right direction. Mr. Mohebbi Holding a Master of Science in Electrical Engineering from George Washington University, he worked as an Associate Professor at the University of Wisconsin before transitioning into the transit industry. With over 30 years of experience, Mr. Mohebbi has focused on industry engineering and product development, contributing to the programming of ERSA and the patented design of the Mobile App Roaming System (MARS) through his expertise in cellular roaming systems.

He has invested in several mid-size ground transportation services and logistics businesses, working closely with their management teams to foster growth and forge strong relationships within the transportation and logistics sectors. Mr. Mohebbi's solid background in mathematics has not only enabled him to secure a multi-year industrial partnership grant with the University of Maryland but also to obtain several FTA innovation grants in partnership with transit agencies across various regions, some of which are currently in progress. He has co-authored numerous papers on the Dial-A-Ride and ridesharing applications of the projects funded by these grants.

Kevin Rivas, Account Manager & Proposal Writer

krivas@itcurves.net, (301) 208-2222, ext. 3256

Mr. Rivas will provide training services, documentation, installation of MDTs into vehicles (if required by IT Curves), and post-delivery support services. As a customer support manager at IT Curves, he holds an associate's degree and has been with the company since 2015.

Mr. Rivas is responsible for user-testing, product development, and customer support and training, focusing specifically on the interaction between in-vehicle equipment and office systems. He is an expert in all areas of IT Curves' products and works closely with the development team on product design and adjustments based on customer requirements.

In addition to his responsibilities in design and device installation, Mr. Rivas led the development of the Passenger Information Module, which offers interactive information and entertainment for passengers in vehicles. He also contributed to the development and testing of the iMeter, a virtual taxi meter that charges based on GPS distance instead of OBD-II connections. Mr. Rivas led the project to attain NTEP certification.

Imran Siddiqui, Chief Technology Officer, Assigned Project Manager

myounus@itcurves.net, (301) 208-2222, ext. 3249

Mr. Siddiqui will serve as the executive advisor for this project team. He will lead the requirement analysis to ensure that the requirements are fully understood and that our team of experts completes the demonstration, installation, and testing processes in compliance with these requirements. With extensive direct experience in our paratransit and route maker software, Mr. Siddiqui will guarantee the correct implementation of every feature. He is responsible for application software design, technical and functional specification gathering, project planning, and real-time production support.

Mr. Siddiqui holds a Master of Computer Science degree, having conducted in-depth studies of advanced databases and algorithm analysis and design. He leads the development of cutting-edge applications that connect customers and vehicles for optimal allocation and resource assignment based on location and the best route. Mr. Siddiqui also worked in the Department of Civil and Environmental Engineering at the University of Maryland, where he helped develop the ERSA and optimize it for various modes of operation. He played a key role in converting this research work for commercial application, which now serves as the engine for our routing module in automated dispatch. Currently, Mr. Siddiqui leads our team of twenty-five engineers in system design and maintenance.

Dave Benton, In-Vehicle Network Design & Integration

dbenton@itcurves.net

In 2023, IT Curves was fortunate to welcome Mr. Benton into its team, an individual whose drive, technical prowess, and unwavering dedication to client success quickly came to the fore. A recent graduate, Mr. Benton completed his Bachelor of Science in Computer Science from the esteemed University of Maryland - Baltimore County, in December of 2022. Throughout his educational journey, Mr. Benton meticulously honed an impressive array of skills. From developing fluency in multiple coding languages to mastering the development for diverse operating systems, his comprehensive capabilities have been of immense value. Further, his expertise in SQL database development and management has further bolstered his formidable technical arsenal.

Since his advent at IT Curves, Mr. Benton has embarked on the intricate task of transitioning existing applications from Android to Linux. This complex process encompasses a myriad of our proprietary features, originally designed for IOS and Android, that are integral to both our driver and rider experiences. His commitment to ensuring seamless adaptation and functionality is a testament to his professional acumen and dedication.

Abhijeet Nawale, Software Developer - Integration & Customization

ajnawale@itcurves.net, (301) 208-2222, ext. 3279

Mr. Nawale holds a Master's degree in Computer Science from Arizona State University and a Bachelor's degree in Computer Science from Vellore Institute of Technology. He is currently collaborating with Florida State University to enhance the ride-sharing algorithm.

In addition, Mr. Nawale maintains the API documentation and ensures the public API remains up-to-date with the latest release by incorporating new features. He has developed a dynamic ride-booking website for customers, a utility tool for dispatchers to convert tabular data into the dispatch system, and contributes to projects requiring integration and customization.

3. Client References

Grant Transit Authority (GTA)

Demonstration of Delivery

Agency Name: Grant Transit Authority (GTA)
Size: Rural Transit Agency
System Provided: Paratransit, Fixed Route, and IRS APC kits

Reference

Contact Name: JoBeth Carlson
Title: Operations Manager
Address: 8392 Westover Blvd
Moses Lake, WA 98837
Phone: (509) 765-0898
Email: jobethc@granttransit.com

Project Summary

IT Curves has recently been awarded a contract by Grant Transit Authority (GTA) to provide an integrated ITS solution for fixed bus and paratransit services in Moses Lake, WA. With GTA operating a fleet of 33 vehicles across their two operations, our solution included a fixed bus route planning and management module, as well as our paratransit fleet management and scheduling modules. GTA has expressed high satisfaction with the services we have provided since the initiation of the project and initial deployment.

To ensure a comprehensive understanding of the system, IT Curves includes in-depth training with every software purchase. As part of the cutover process, our delivery team installed Mobile Data Terminals (MDTs), provided on-site training for their configured system, and were on-site for the installation of the IRIS APC kits. We also scheduled online meetings via zoom, where GTA staff were able to ask any questions and receive status updates throughout the cutover process.

In addition to the training sessions, we provided an easy-to-follow user manual for their system and continually provide 24/7 technical support. We also provided training for county technicians on how to install the driver MDTs into the vehicles, should the need for arise.

Allegany County Transit

The projects below consist of similar projects that have been delivered to transit agencies and transportation providers with a similar size and scope as the individual agencies listed.

Allegany County Transit

Demonstration of Delivery

Agency Name: Allegany Count Transit
Size: Rural Transit Agency
System Provided: System for Paratransit, Fixed Route Service, and IRIS APC Kits

Reference

Contact Name: Elizabeth Harper
Title: Transit Chief
Address: 1000 Lafayette Ave
Cumberland, MD 21502
Phone Number: (301) 722-6360
Email: erobison-harper@alleganygov.org

Project Summary:

IT Curves has successfully provided technology solutions for Allegany County Transit, a paratransit and fixed bus route service provider in Maryland. Allegany operates a fleet of 20 vehicles across their two services.

We supplied them with our fixed bus route module, a configured version of our paratransit dispatch and scheduling modules, and IRIS APC kits for their fixed route vehicles. They have expressed great satisfaction with the transit management solutions we have delivered.

As part of our comprehensive service, IT Curves provided in-depth training to ensure Allegany County Transit staff fully understood the system. This included an easy-to-follow user manual, onsite training, and 24/7 technical support. We also trained county technicians on installing driver MDTs in their vehicles and demonstrated the process on approximately half of the fleet's vehicles.

Albany Transit System

Albany Transit System (ATS)

Demonstration of Delivery

Agency Name: Albany Transit System
Size: Rural Transit Agency
System Provided: System for Paratransit

Reference

Contact Name: Luke Cotton
Title: IT Project Manager
Address: 333 Broadalbin St SW
Albany, OR 97321
Phone Number: (541) 917-7538
Email: luke.cotton@cityofalbany.net

Project Summary

IT Curves successfully won a highly competitive RFP to deliver Paratransit software to the Albany Transit System. The project was delivered in 2022 and involved fulfilling various requirements, including a cloud hosted paratransit package, scheduling and routing tools, dispatching software, funding agency reporting, MDT application, and continuous support. We proudly met and exceeded each requirement, particularly in terms of our customer support. So much so that Albany Transit decided to purchase optional add-on packages after initial deployment.

The add-on package that Albany Transit purchased and we delivered was the Smart Telephony System. This system included an integrated Interactive Voice Response (IVR) platform that allows riders to interact with the Paratransit system via Phone, check the status of their trip, and receive notifications about their ride. This feature ensures riders are informed of the status of their ride at every step, including receiving scheduled vehicle information the day before and real-time arrival reminders.

Our customer support sets us apart from the competition. We are dedicated to resolving major issues within 24-36 hours and minor issues within just 4 hours. Our responsiveness approach and commitment to closely working with our customers are key factors in our clients becoming returning customers.

4. Overall Approach to the Scope of Services

4.1 Scope of Work

1. Route Planning/Mapping

1.1 Route planning and mapping should provide GTA users with tools to create and modify routes on a commercially available/open-source mapping platform that will be reflected in the timetable.

Response: IT Curves will provide GTA with the Route Planning tools needed to plan, create, edit, and optimize routes in the software. Users will have various tools available to them, which they can use to make desired changes as needed. If GTA requires changes on existing CAD/AVL tools, IT Curves will work with GTA staff to improve existing tools as well as add new tools to the software.

Refer to Appendix for proposed ideas.

1.2 GTA users should be able to create bus stops/stations on the map and the ability to upload existing stops into the software using GTFS data. Bus stop stations should be differentiated by timepoints, non-timepoints, depot, and operator relief points.

Response:

i. IT Curves current CAD/AVL provides users with the ability to create and edit new stops directly from the Route Templating tool. The tool provides GTA staff with a simple way to create new stops in designated locations by simply clicking the map. Alternately, GTA staff can type an address to add it into the software.
ii. Stops can be color coded and named according to the needs of the operation. These options can aid in quickly recognizing and or filtering stops.
iii. Stops can be categories as on-demand stops or scheduled stops, depots and transit centers. It is easy to add new stop types i.e., relief points and that can be later used for reporting purposes
iv. IT Curves current CAD/AVL contains configuration utilities which are utilized by IT Curves deployment team to upload stop and route information using GTFS online or predownloaded files. IT Curves will be making it part of user accessible features.
v. The GTFS upload tool will be integrated and available to the GTA team at no additional cost.

1.3 *Existing routes should be able to be modified within the map to finetune the route or examine other possibilities including future alternatives.*

Response: IT Curves current CAD/AVL system does allow modifying the stops and driving route between two stops within the route segment. IT Curves plan to enhance the user interface to allow GTA planning team to modify stops along the route directly from the map. Staff will have the ability to click a stop and move it along the map, visually see how the stop change will affect the route path, then apply this change to the route. The change will facilitate and improve the user experience while modifying existing routes in the software.

1.4 *The GTA user should have the ability to customize route names, route patterns, customize route colors, identify day of week operations, appropriate vehicle type, and other standard transit route planning features.*

Response:

i. IT Curves current CAD/AVL provides GTA staff with the ability to create and edit routes directly from the Route Configuration module. GTA staff can create custom routes using desired naming convention, color, codes, selecting operational days of the week, and start & end times. The current screen allows users to create a limitless number of routes with custom configurations as needed for operation purposes.

ii. IT Curves will improve the current tool by adding the option to attach attributes to the routes. Attributes will allow for further customization of routes. Staff will be able to create a limitless number of attributes including the required vehicle type. Attributes will be used in finding the best vehicle for the routes. Furthermore, staff will have the ability to narrow down reports by attributes attached to both the vehicles and the routes.

iii. IT Curves will incorporate the use of machine learning in the software to provide more efficient routes based on the information available from GPS data points of vehicles.

1.5 The route planning module should use Google Maps, Apple Maps, Open Street Maps or equivalent for a platform to offer geocoding and layering.

Response:

i. IT Curves software utilizes Google Maps to present data to GTA Staff. The Planning tools, stop creation and plotting, are all visualized on the map and are readily available to all staff.

ii. Planning engine utilizes the Open Street maps for additional layers and adding driving way points.

iii. Furthermore, AVL data is presented in Google Maps, both the live tracking and the past playback.

iv. IT Curves is planning to add additional layering features i.e., creating barriers and roadblocks in the system. This will assist the route planning team to know if any detours are needed while planning or for already planned routes.

1.6 Demographic layers with US Census data are a required component for the planning software. Demographics should be customizable by the GTA user using official census data or surveys with different metrics for current routes and future planning.

Response: IT Curves currently has separate tool designed for enhancing and planning fixed transit service by introducing new stops by utilizing multi-dimensional census and service data i.e.,

i. Census data at tract and block level from publicly available APIs
ii. Proposed new stops to add more convenience, increase coverage and reduce operation costs.
iii. Determine existing less frequently utilized stops for onboarding and alighting by uploading passenger counter data directly from the APC or user uploaded files.
iv. Recommend converting fixed stops to OnDemand stops and reduces total trip length.
v. will provide GTA with tools to visualize Demographic data within the map for planning purposes. GTA staff will be able to plot routes and utilize demographic data to modify routes around those parameters, to better serve customers and or reduce deadheads. Staff will be able to select from various parameters to compare statistics and improve routes.

IT Curves will integrate this tool into the current CAD/AVL software available to GTA planning staff. Staff will be able visualize where the population density lies within the city to plan route stops and hubs. The data directly pulled from the US census.

IT Curves will design and implement as an integrated feature in response to this proposal – free feature enhancement

1.7 Software should allow for current GTA ridership statistics to be integrated to enhance future planning and identify passenger boarding and alighting needs. It is desirable to determine radius distances from each stop for various intervals (i.e. ½ mile, ¾ mile, etc.)

Response:

i. IT Curves service planning enhancement tool integrates with the ridership using APC or uploaded spreadsheets as extensive reporting tools are available which are available to provide stop by stop ridership. Both tools together allow the planning team to visually determine which stops have heavy ridership and helps in considering those stops schedule as timed stop for future planning.
ii. All the stops are also available on the map along the route, IT Curves will enhance the current map tool to calculate the distance of any location on the map from the nearby planned stops.

A. Timetable Planning

i. *Timetable planning should provide GTA users with the tools required to build a service schedule data set. The software should enable GTA users to define a route from scratch by selecting the route stops, defining time points, setting the distance between the stops/time points, etc.*

Response: IT Curves current CAD/AVL system includes the required feature.

ii. *The user should be able to define the following (manually or by uploading from Excel):*

Response: IT Curves current CAD/AVL system allows defining stops manually. IT Curves will enhance current CAD/AVL to allow uploading stops file and GTFS for other information as desired below.

a. *Different route services for weekdays, weekends, holidays, and any other special service*

Response: IT Curves current CAD/AVL system includes the required feature. Users can define separate route patterns for weekdays, weekends, holidays and special events and service days.

b. *Route patterns*

Response: IT Curves current CAD/AVL system includes the required feature.

c. *Allowable vehicle types for each of the route patterns*

Response: IT Curves will enhance current CAD/AVL to associate desired vehicle types for each route segment/patterns.

IT Curves will design and implement as an integrated feature in response to this proposal - Priced feature

d. *Travel time between stops for different time bands of the day*

Response: IT Curves current CAD/AVL system includes the required feature. Planning teams can create different segments for hours of day and can define different travel patterns, time and distance between stops.

e. *Start time, end time, and frequency for each route*

Response: IT Curves current CAD/AVL system includes the required feature.

iii. *Based on the data above, the system should generate a full timetable for any given route. The GTA user should have the ability to later make any manual amendments to the plan, with the system automatically updating the subsequent stop times.*

Response: IT Curves current CAD/AVL system includes the required feature.

2. Scheduling/Runcutting

2.1 GTA requires a scheduling platform as part of the software suite. The scheduling software should allow GTA users to cut runs and assign vehicles to routes from the created routes under "Route Planning/Mapping."

Response: IT Curves current CAD/AVL system allows creating route segments (also known as trip variations). Then route segments are combined to create route configuration including the variations i.e., for specific days of weeks. Route configuration auto generates the block sheets with defined headway and vehicle blocks. The block/sheet in form of time chart allows to further modify each stop

a. Schedule arrival time
b. Converting stop as OnDemand
c. Excluding the stop from trip

*2.2 The scheduling platform should provide the tools to easily create a fully optimizable vehicle and crew schedule and compare multiple "what-if" scenarios, using clear and accurate key performance indicators (KPIs)***.**

Response: IT Curves current CAD/AVL system allows combing route segments into the runs. These runs can further assign drivers for different days of the week. While drivers are being assigned to the runs, the system keeps track of driver weekly hours and prompting user for current week hours.

The current CAD/AVL system does not provide the ability to compare different models but IT Curves will work with GTA to comply with their requirement and enhance system where different models can be compared with each other.

IT Curves will design and implement as an integrated feature in response to this proposal - Priced feature

2.3 The GTA user should be able to customize preferences and constraints including, but not limited to vehicles (including alternative-fueled vehicles), operators, depots, and federal, state, and local policy/procedure restrictions, including specific requirements from local collective bargaining agreements. The scheduling platform should give the GTA user the ability to fully optimize vehicle and operator schedules in a matter of minutes within the specified requirements. The user should also be able to make any manual run cuts, including amendments, changes in preference parameters, and optimize multiple times to produce additional scenarios. KPIs such as total cost, paid time, overtime, total miles, total hours, etc. should be comparable for each optimization performed.

Response: IT Curves attribute system allows to classify vehicle type and other requirements at route segment/ trip level. IT Curves will enhance the system to associate cost with each resource allocated to runs and provide a tool to compare different models. IT Curves optimization tools will calculate cost based on the

a. driving distances including deadheads between routes
b. vehicle operating cost
c. total paid time and overtime in case of extended hours.

IT Curves will design and implement as an integrated feature in response to this proposal - Priced feature

3. ***Employee Scheduling/Rostering***

3.1 GTA requires the software suite to include a rostering platform to use the duties schedule created in the scheduling/runcutting platform to produce a daily or weekly (or any other working cycle) roster. The rostering platform should have a set of preferences that are defined to capture any collective bargaining agreement rules such as roster types, days off, overtime, cost parameters, etc. The GTA user should be able to review any discrepancies and compare multiple scenarios by predefined KPIs and overall cost.

Response:

a. IT Curves CAD/AVL system currently provides an easy and flexible way of assigning drivers to Runs. While driver assignment, the current tool Keeps track of total weekly hours of assigned driver and allows to make temp assignments during the days off.
b. New feature of attributes assignment will be added to the CAD/AVL which will enable the planner to assign runs to the driver with matching rules.
c. The current system allows to define each employee weekly hours and overtime rules. IT Curves will enhance the CAD/AVL to integrate with employee. management system to bring in work hours and defined days off for each driver.
d. A route planner will be able to create multiple rostering models and compare them for cost optimization based on driver hours and overtime.

4. **User Profiles & Training**

4.1 The Contractor will provide GTA with at least two administrative user profiles and six viewer user profiles. The Contractor will provide GTA staff with adequate initial and ongoing training at the GTA's discretion, including new employees hired during the term of the contract.

Response: IT Curves current CAD/AVL allows GTA to define an unlimited number of users. Each user can be assigned different access levels within the system, and access can be restricted based on assigned roles. User roles determine module-level access, ensuring that only authorized users can access employee profiles. Furthermore, only users with higher-level permissions can view sensitive personal information, such as Social Security numbers and driver's licenses.

5. **Reports & Software Integration**

5.1 GTA requires the scheduling platform to generate a set of predefined built-in reports for driver/block schedules, usage and metrics, route summary, revenue trip summary, and others. Heat maps or similar reports for passenger boarding/alighting is also desired.

Response: IT Curves is committed to meeting GTA's reporting needs by leveraging our existing tools and resources. Currently, our suite includes the Passenger Load Report and Demographic Report, which provide insights into customer counts and demographic details, respectively. In response to GTA's requirement for predefined built-in reports, our software will be enhanced to include:

a. **Driver/Block Schedules**: We will generate reports that detail driver assignments and block schedules.
b. **Usage and Metrics**: Reports will track various usage metrics to help in operational analysis.
c. **Route Summary**: Detailed summaries of route performance will be provided.
d. **Revenue Trip Summary**: Reports will focus on revenue generated from trips.
e. **Passenger Boarding/Alighting**: We will incorporate heat maps or similar visual reports to illustrate passenger boarding and alighting patterns.

5.2 Rosters should be in a format customized to GTA's bidding process, or an agreed upon format within the software's capabilities.

Response: To address the need for rosters customized to GTA's bidding process, IT Curves is prepared to adjust the format of the rosters within the capabilities of our software. We will work closely with GTA to ensure that the roster format meets your specific requirements or aligns with an agreed-upon standard.

4.2 IT Curves' ability to Complete the Project in a Timely Fashion

IT Curves will provide an integrated Route Scheduling/Runcutting and Dispatch Planning Software System for Scheduling/Runcutting and planning as per the description of the RFP. The software will be fully integrated with current CAD/AVL and dispatch system already used by Grant Transit Authority. To provide this system, IT Curves will perform the following tasks:

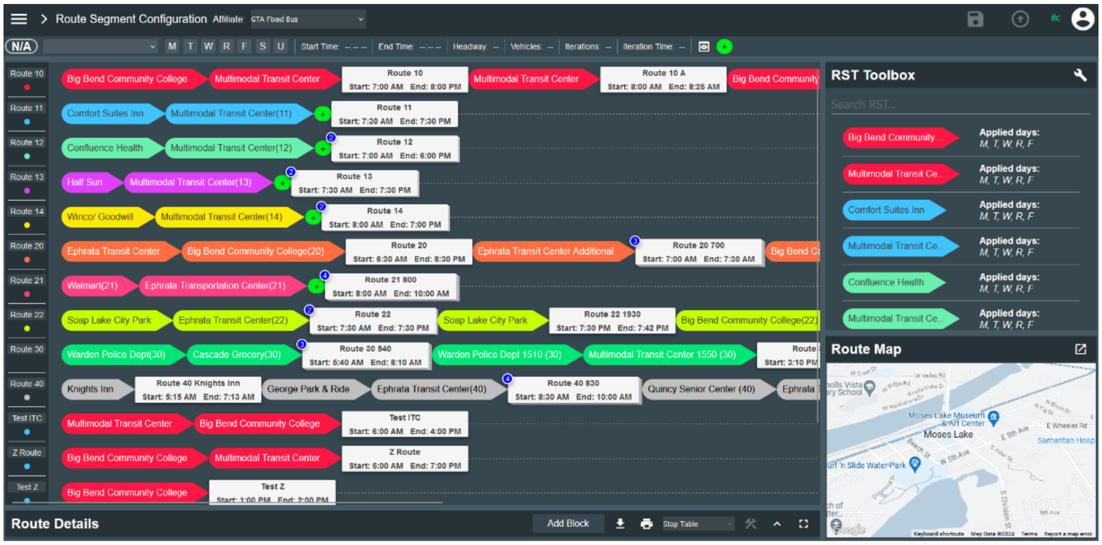
1. Supply proposed products
    a. Fixed Route Scheduling System: IT Curves will provide our enhanced fixed route Scheduling/Runcutting and Rostering module integrated into the current deployed MRMS software package which currently provides solution for both fixed route and Demand Response reservation, scheduling, and dispatch.
2. As the newly offered software module will be fully integrated into the current CAD/AVL system, there won't be any additional data needed from GTA. IT Curves will be able utilize data directly from current CAD/AVL system. The new module will be needing routes block sheet or can download it through GTFS for initial setup. If any data needs to be transferred, the method of transfer, data to be provided, and GTA involvement will be determined during the discovery phase.
3. Conduct detailed training of GTA staff: As detailed in Training and Documentation, IT Curves' installation schedule includes a thorough training program for GTA dispatch staff and drivers.
4. Provide manuals and other training resources for GTA: As detailed in in person Training and Documentation, IT Curves will provide GTA with training resources and manuals after the training period of the Implementation schedule. Our staff remain available to provide additional training sessions upon request, with pricing noted in the Price Proposal.
5. Ensure successful and timely implementation: As detailed in System Installation and Implementation, IT Curves anticipates a nine-week schedule to completely transition GTA to MRMS.
6. Provide ongoing support for GTA: As detailed in Maintenance and Warranty Plan, IT Curves provides ongoing support for all clients, including not only emergency support, but additional training, technical support, and updates to the software to ensure its ongoing performance and functionality.

4.3 Technical Proposal

Route Planning/Mapping

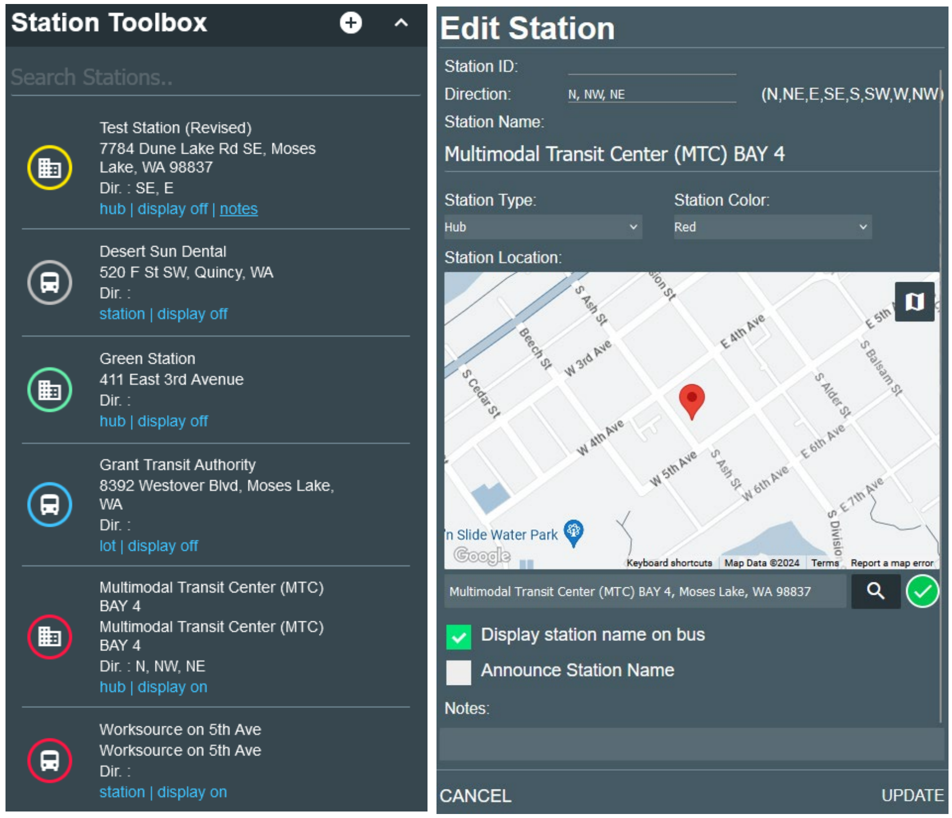
1.1 IT Curves is currently providing GTA with tools for route creation and modification. These tools include the Station Toolbox, Edit Station, Route Segments Templating, Route Configurations, and Route Blocking.

IT Curves will be making improvements to the existing modules and introducing new ones, such as Demographic Route Planning.

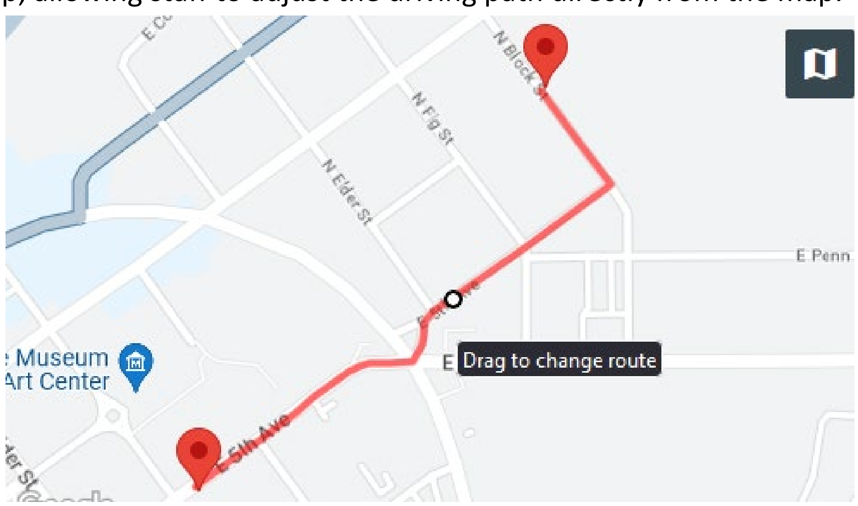


1.2 The Station Toolbox and the Edit Station tools allow users to create and edit stations within the software. In the Station Toolbox, staff can view which stations are currently available in the system. Stations can be edited, deleted, or new ones added by clicking the "+” button at the top of the list. The system allows to assign the color to stations, there are different types stations system allows to define “Hub”, “Virtual Stop”, “On Request” and “Lot”, it also allows assigning the approaching direction of the station as an attribute of definition.

IT Curves will improve the Station Toolbox by adding an option to search by station type and color.

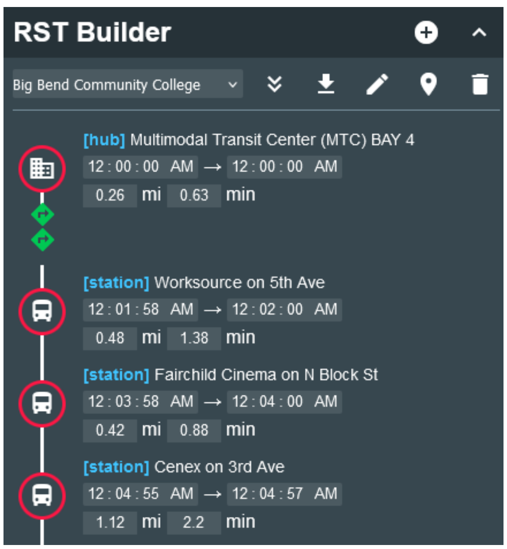


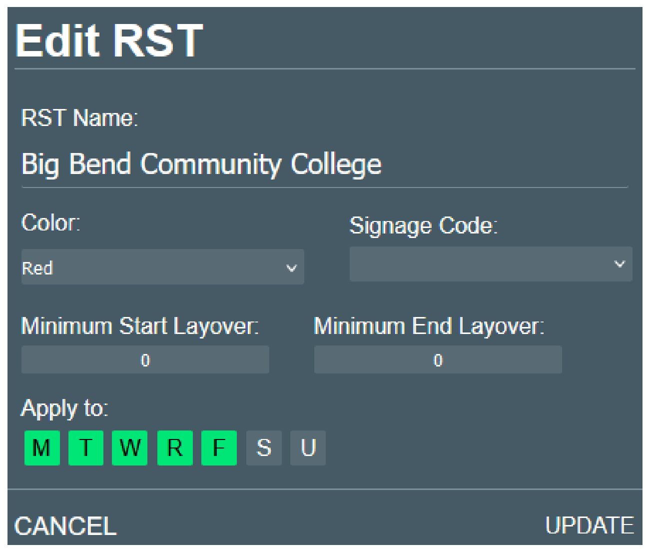
1.3 IT Curves will enable staff to click on the driving path on the map and adjust it based on their selections. Currently, the system allows modifications to the path between two stations. In the new implementation, the software will display the entire route on the map, allowing staff to adjust the driving path directly from the map.



1.4 IT Curves provides staff with the ability to quickly create or edit route segments using the RST Builder toolbox. Staff can name the route segment, assign a color, and select the days of the week the segment is available for routes. By dragging and dropping stations into the list, routes are created, and for each station added, the map displays a station pin and the corresponding driving path. Time intervals between stations and distances can be adjusted to fine-tune information for reporting and for display to customers. The information entered in the RST Builder can then be used to create routes and the blocking schedule.

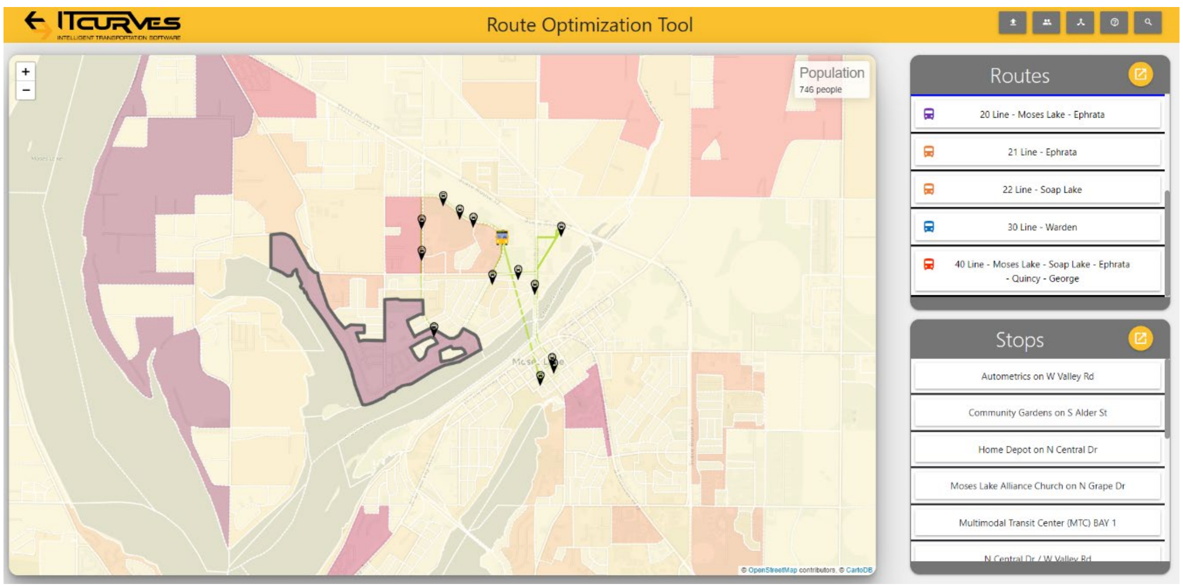
IT Curves will enhance this tool by allowing users to adjust routes directly on the map screen. As staff move stations to different locations or modify the driving path, the toolbox will automatically update the times and distances to reflect the latest changes.

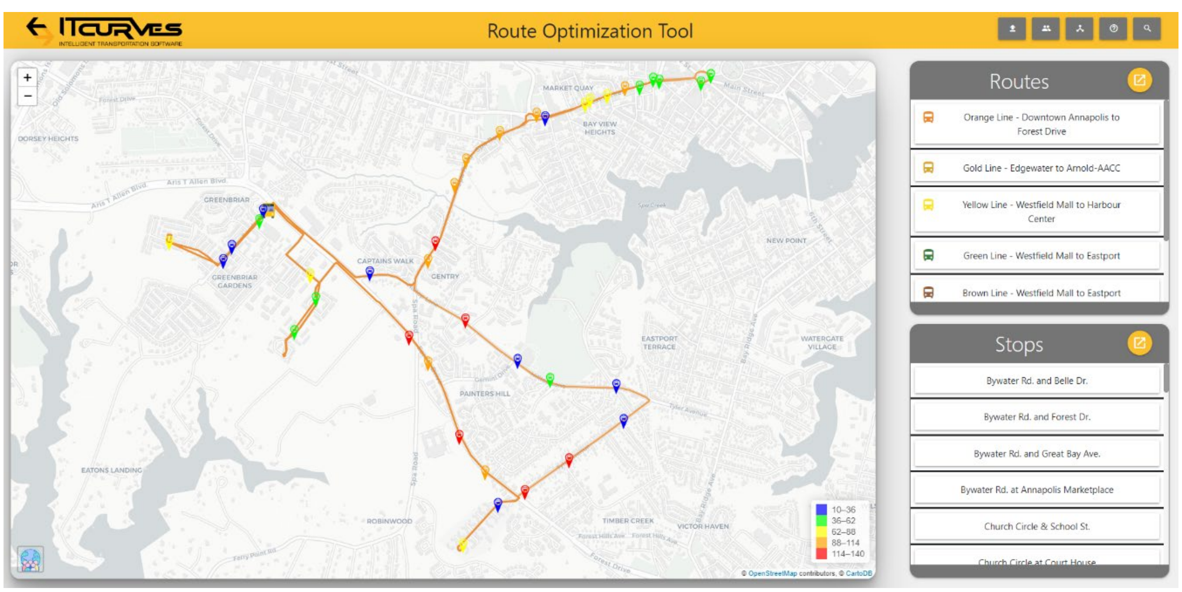




1.5 IT Curves will provide GTA with a Demographic Tool that contains information about the current population size in their region. The tool retrieves data from the census and breaks it down into blocks. Areas are shaded to represent population densities, with lighter colors indicating lower populations and darker colors indicating higher populations. This information is valuable for planning routes and stations to better serve customers across the region.

Additionally, the tool can integrate APC (Automatic Passenger Counter) data to visualize frequently visited stations, using red to indicate more frequently visited stations and blue to indicate less frequently visited ones. Staff can filter the data by selecting a specific time of day, allowing them to view APC data for that time range, which helps in identifying the most popular stations during different periods.





1.6 IT Curves provides staff with ridership data through Demographic and Passenger Load reports. The demographic report breaks down data by runs, with information pulled directly from devices during the route life cycle. Similarly, the Passenger Load report includes data from the route during its run, provided by the Automatic Passenger Counters installed in the buses. These reports display the number of passengers boarding and exiting the vehicle.

Staff can filter the reports by vehicle, route, or run, and further sort the data by stations, routes, or time.

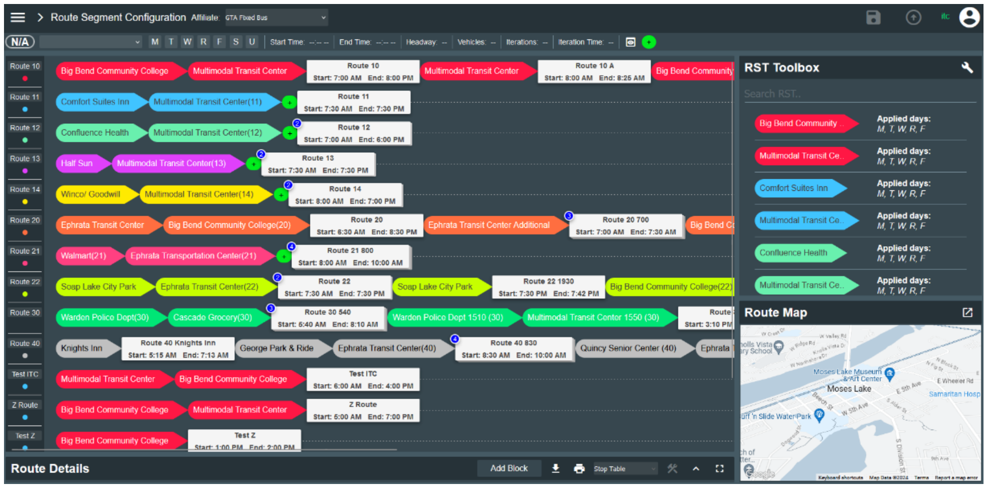
Passengers Count Report (Summary)								
	Driver Device		Door 1		Door 2		Total	
Route	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	0	0	45	62	0	0	45	62
Route 10	96	0	137	111	2	35	139	146
Route 11	46	0	47	49	0	1	47	50
Route 12	37	0	23	20	0	0	23	20
Route 13	108	0	24	27	0	0	24	27
Route 14	46	0	51	47	0	3	51	50
Route 20	62	0	93	93	0	6	93	99
Route 21	6	0	8	9	0	1	8	10
Route 22	37	0	32	19	0	6	32	25
Route 30	20	0	43	38	0	0	43	38
Route 40	28	0	28	18	0	5	28	23
Test Z30	0	0	0	0	0	0	0	0
						Total:	533	550

RUN: B			Driver: Elias Jara			Bus#: 2604			Date: 08/20/2024			
Route#	Start Time	End Time	Actual Time	End Time	Children	Youth	Adult	Senior	Total	Bikes	Service Animal	Wheelchair
Route 10	08:00:00	13:09:00	07:59:45	13:48:00	0	0	13	4	17	1	2	0
Route 11	09:30:00	09:49:58	09:31:31	09:48:45	0	0	3	0	3	0	0	0
Route 13	09:00:00	09:24:58	09:01:15	09:20:30	0	0	0	0	0	0	0	0
Route 14	10:00:00	10:49:58	09:59:05	10:52:45	0	1	8	3	12	0	0	0
Route 30	05:32:00	07:59:58	05:33:00	07:58:30	0	0	6	3	9	0	0	0
Total:					0	1	30	10	41	1	2	0

Timetable Planning

IT Curves software allows users to create unlimited Route Segments and Route Configurations. Staff will have the ability to fully customize segments, including stations, times, visuals, and availability throughout the week. These segments can then be used to create routes by simply dragging and dropping them into routes. Staff can also add configurations to control the dates and times the routes will be available.

IT Curves will enhance route planning by introducing Route Attributes. These attributes can be created and assigned to routes, drivers, and vehicles to ensure compatibility during the assignment process.



Lamb Weston on E St SW	Knights Inn on F St SW	Quincy Inn & Suites on F ...	Akins Fresh Foods on F ...	Andaluz family Mexican ...	Quincy Community...	Jakes Meats on Hwy 281	Shree's Truck Stop & Gas...	Shree's Truck Stop & Gas...	L&R Cafe on Central Ave S	Fred Lischka Properties o...	F St SE / 1st Ave SE	Jackpot Food Mart on F St...	Lamb Weston on E St SW	Quincy Senior Center on F
On Request	08:53:00 A	08:53:31 A	08:54:06 A	08:54:43 A	08:55:12 A	On Request	09:15:00 A	09:15:00 A	09:30:12 A	09:30:29 A	09:30:52 A	09:31:14 A	excluded	09:35:00 A
On Request	01:23:00 P	01:23:31 P	01:24:06 P	01:24:43 P	01:25:12 P	On Request	01:45:00 P	01:45:00 P	02:00:12 P	02:00:29 P	02:00:52 P	02:01:14 P	On Request	02:05:00 P
On Request	On Request	03:23:31 P	03:24:06 P	03:24:43 P	03:25:12 P	On Request	03:45:00 P	03:45:00 P	04:00:12 P	04:00:29 P	04:00:52 P	04:01:14 P	04:03:01 P	04:05:00 P
On Request	05:23:00 P	05:23:31 P	05:24:05 P	05:24:43 P	05:25:12 P	On Request	05:45:00 P	05:45:00 P	06:00:12 P	06:00:29 P	06:00:53 P	06:01:14 P	On Request	06:05:00 P

Scheduling/Runcutting

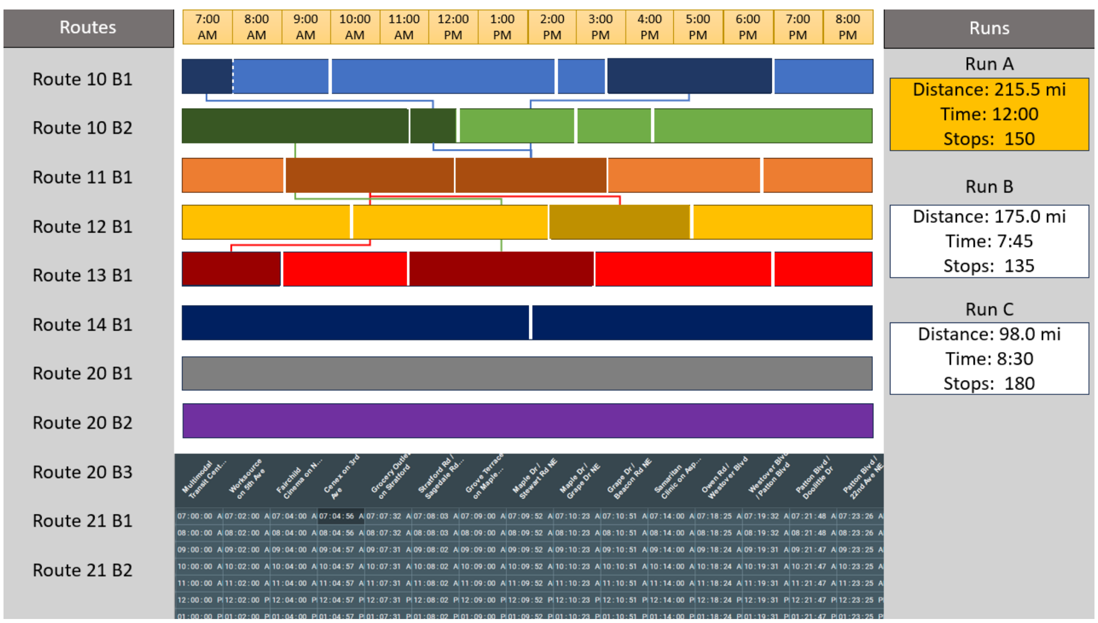
IT Curves software enables staff to use the Run Cutting and Shift Scheduling screens to assign routes to vehicles and drivers. The Run Cutting screen allows staff to divide a route into multiple runs, creating an unlimited number of runs with various route and day assignments. When a route block is assigned to a run, the screen displays the run name and the selected day of the week.

In the Shift Scheduling screen, runs can be assigned to drivers or vehicles. Staff can select the runs and days of the week they want to assign to a driver or vehicle. As assignments are made, the screen displays the details, including the number of hours per day and per week each assigned driver will be driving.

If staff need to make changes to the schedule for any day—such as when a driver calls in sick—they can use the Daily Dispatch screen. This screen allows for changes to be made for the current day only.

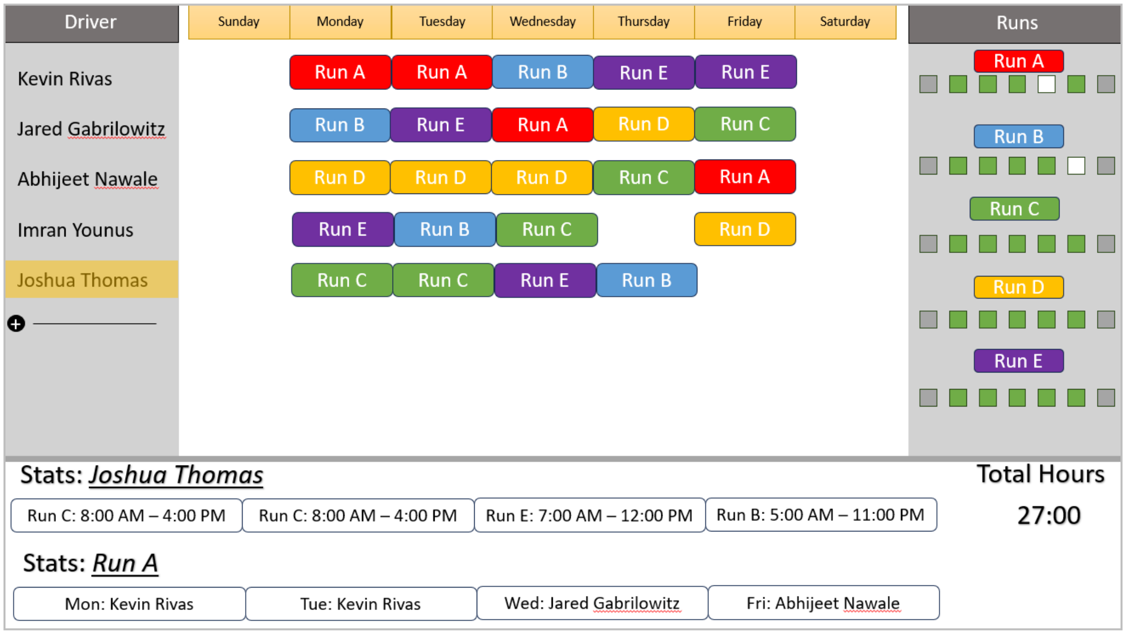
IT Curves will enhance the run cutting and driver assignment processes by adding more visual components to the screen. Staff will be able to break down routes directly on the screen and combine multiple routes to create a run. To split a route, simply click on it and select the option to divide the route. The newly created segment can then be assigned to an existing run or used to start a new run. The run can be expanded to show stations and times. When details are expanded, staff can mark portions of the route as On Request stops, Inactive, or add break points for drivers.

As staff select routes, a clear indicator will show that the route is already in use within a run. The route segments will change to a darker shade to warn staff of this, though they will still have the option to use the segment for a different run if needed. Staff will also immediately see the total hours, miles, and stops the runs will cover based on the selected segments. Runs can be assigned to different days of the week by clicking on the run to open the configuration page, where staff can assign the days and dates the run will be available.



Employee Scheduling and Rostering

The driver/vehicle assignment screen will allow staff to assign the different runs created in the run cutting module to drivers. Drivers will appear on the left-hand side, selected from a dropdown list of active vehicles/drivers. Staff can click and drag the runs to a day of the week to assign them, with the block visually appearing in the row. Staff can highlight a driver or run to view a summary: for drivers, it will display the assigned runs, schedule, and total working hours for the week; for runs, it will show a summary of the drivers assigned. A quick summary of the days of the week that the run has been assigned will also be visible on the right side of the table. Staff will have the option to print or download the driver schedule to share with drivers and other office staff.



WORK SCHEDULE

September, 2 2024 THRU September, 6 2024

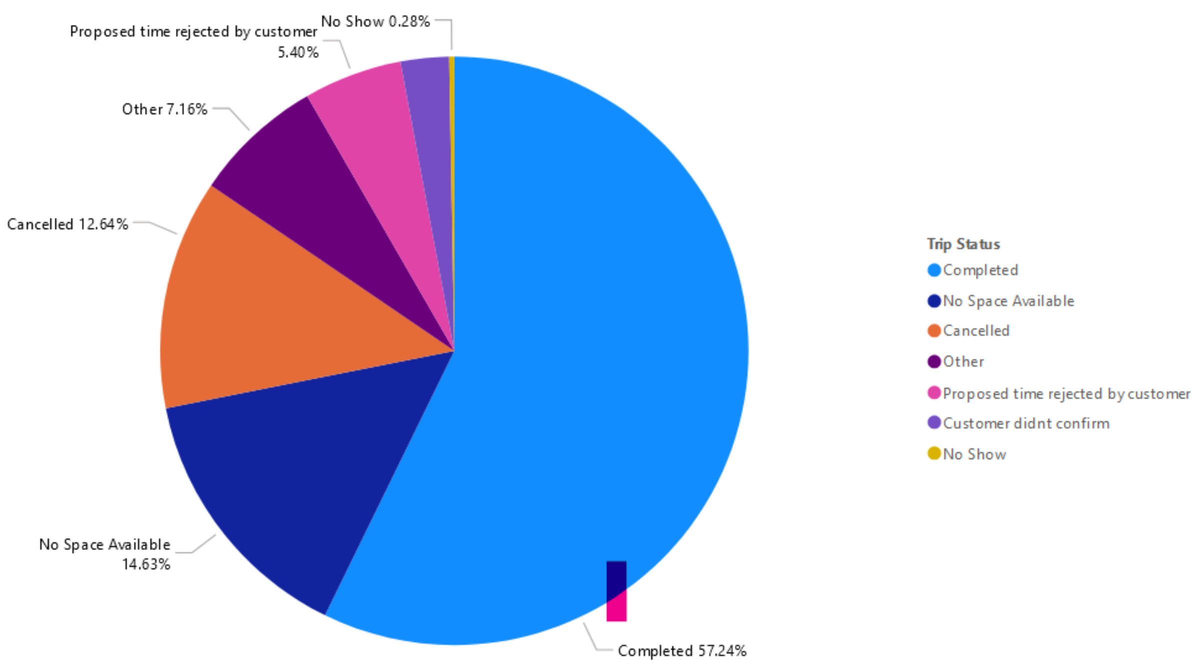
Routes	Run A	Run B	Run C	Run D	Run E	Run F	Run G
Route Time	5:15 AM - 12:30 PM	5:40 AM - 1:00 PM	6:30 AM - 11:30 AM	7:00 AM - 11:00 AM	7:00 AM - 2:00 PM	7:00 AM - 1:30 PM	7:30 AM - 2:30 PM
LUNCH							
Route Days	M,T,W,R,F	M,T,W,R,F	M,T,W,R,F	M,T,W,R,F	M,T,W,R,F	M,T,W,R,F	M,T,W,R,F
DAY & DATE							
MON 2	Andrey Voysekhovskiy	Elias Jara	Gloria Cornish	Fidel Barajas	Lee Alvarado	Michael Chesnakov	Melody Faircloth
TUE 3	Michael Chesnakov	Elias Jara	Jim Ackley	Lee Alvarado	Melody Faircloth	Andrey Voysekhovskiy	Fidel Barajas
WED 4	Michael Chesnakov	Fidel Barajas	Andrey Voysekhovskiy	Melody Faircloth	Lee Alvarado	Elias Jara	Jim Ackley
THU 5	Michael Chesnakov	Lee Alvarado	Elias Jara	Curtis Collins	Fidel Barajas	Andrey Voysekhovskiy	Melody Faircloth
FRI 6	Andrey Voysekhovskiy	Melody Faircloth	Michael Chesnakov		Fidel Barajas	Elias Jara	Lee Alvarado
SAT 7							
SUN 8							

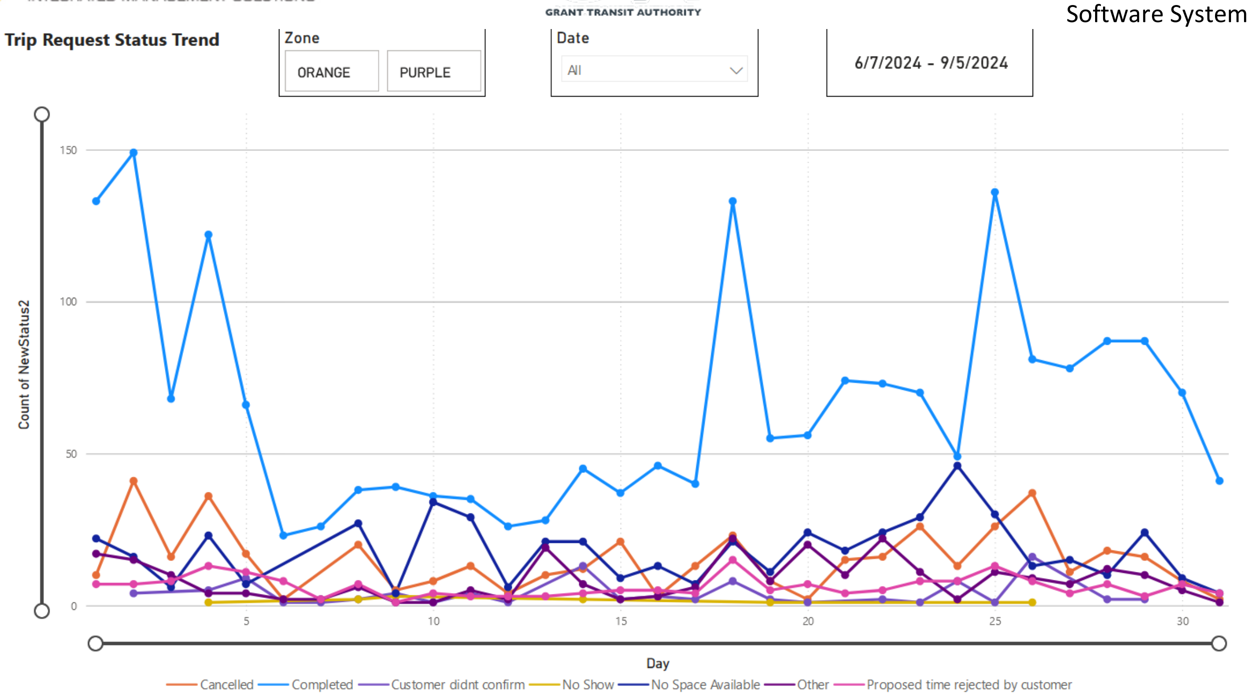
Reports

IT Curves is currently providing GTA with tools for route creation and modification, as well as reports to track various operational metrics. These reports include the Passenger Load Report, which uses APC data to show customer counts on the bus, and the Demographic Report, which provides insights into customer age ranges, mobility types, and more. All demographic data points can be customized to meet specific operational needs. The software also includes additional reports such as driver login information, no-shows and cancellations, route analytics, and more.

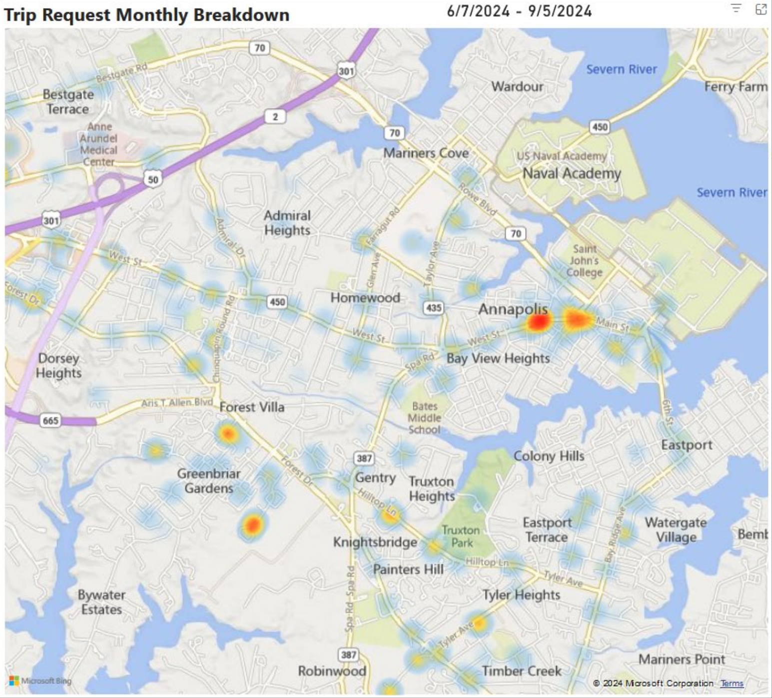
IT Curves will continue to collaborate with GTA to enhance existing reports and develop new ones. As part of these improvements, IT Curves will provide access to Power BI reports, which can be customized to offer graphical statistics. These reports will include trend data, monthly breakdowns, heat maps, and passenger data.

Trip Request Status Overview





Trip Request Monthly Breakdown 6/7/2024 - 9/5/2024



Date

All

Zone

ORANGE | PURPLE

Top 30

Address	DO	PU	Total
Westfield Maill	111	198	309
West St. and Calvert St.	131	145	276
Bywater Rd. at Annapolis Marketplace	126	109	235
West St. and Lafayette Ave.	138	52	190
Newtowne Drive	84	98	182
Bay Ridge Shopping Center	70	107	177
Copeland Street	68	100	168
Social Services	53	96	149
Monroe St. and Bay Ridge Ave.	68	59	127
Newtown	47	76	123
Tyler Ave. and Janwall St.	64	56	120
Robinwood Circle	65	52	117
Tyler Avenue	52	64	116
Main Street / City Dock	48	48	96
West St. and Church Circle	67	21	88
Madison St. and President St.	53	34	87
Festival at Riva	21	59	80
Hilltop Ln. and Boxwood Rd.	38	41	79
Forest Dr. and Harker Pl.	34	39	73
Chinquapin Round Rd. and Drew St.	24	43	67
Bywater Rd. at Annapolis Marketplace , Annapolis, MD 21403	45	19	64
1214 Newtowne Dr	25	34	59
1997 Annapolis Exchange, Annapolis, MD	7	50	57
Total	**1649**	**1773**	**3422**

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4.4 System Delivery and Testing

Our usual schedule for installation, training, and cutover takes nine weeks, and IT Curves always works with our clients to determine the ideal schedule for their needs. Each client's instance of MRMS is installed onto the IT Curves cloud and requires little to no effort from client staff. IT Curves will work on improving the current system while introducing new features and tools. In the developing process, IT Curves will periodically report changes to GTA and will receive feedback for tool enhancements.

As part of our regular delivery planning and process, IT Curves activities include:

a. Work with customers to collect detailed operations requirements and daily practices.
b. For demographic based planning, IT Curves will work with GTA to understand the demographic requirements and rules based on different population-based demographics.
c. IT Curves will work with GTA to capture all operational cost parameters for the route planning tool. Based on this information, the costing tool will enable GTA to evaluate various route options and their associated costs.
d. IT Curves will propose the Run Cutting screen change and gather feedback from GTA to develop, and apply improvements to the tool and enhance its ease of use for staff.
e. Similarly, IT Curves will gather feedback to improve the design and usability of the Driver Assignment screen.
f. IT Curves will collect requirements to include all necessary data points for GTA in the Power BI reports. Additionally, IT Curves will provide different available templates for data presentation.
g. IT Curves will present the driver schedule printout to GTA and make any required changes to the template.

Customer Training and Documentation

As part of the installation process, IT Curves will prepare and deliver a training plan for GTA staff, including both office staff and drivers. The training will include Zoom classes with audio and visual aids, manuals, and other training materials for continued use by County staff. Access to our trainers is part of IT Curves' ongoing maintenance and support plan for the transit authority. IT Curves plans to deliver extensive systems documentation during the discovery and installation processes, including systems testing guidelines, core design specifications, and more.

Information Security Plan

The three information security principles of Confidentiality, Integrity, and Availability, also known as the CIA triad, form the cornerstone of IT Curves' security infrastructure. They function as objectives for the engineering of the IT Curves system, and we are constantly monitoring measuring ourselves on how to better reach these ideals.

The CIA triad is foundational to our information security program. Any time data is leaked, a system attacked, a user takes phishing bait, an account hijacked, a website maliciously taken down, or any number of other security incidents occur, this means one or more of these principles have been violated. No matter how minor the incident, each security breach is taken seriously and addressed as if it were an attack on our entire foundation.

IT Curves' security team constantly evaluates threats and vulnerabilities based on the potential impact they have on the confidentiality, integrity, and availability of our services, applications, and data. Based on that evaluation, the security team implements a set of security controls to reduce risk within the environment.

IT Curves is committed to providing high-availability, secure, and reliable service for our clients. Our network engineering team maintains all processes, performs regular maintenance and testing, and is widely available to address our client's questions and concerns. IT Curves and our individual team members are proud of our record in mitigating risk, responding quickly to client needs, and providing a superior service.

Availability

IT Curves offers MRMS as a high-availability software-as-a-service on our self-owned and maintained cloud servers. We guarantee 99.95% uptime, with all interruptions to service scheduled ahead of time with the client to occur during periods of inactivity or low-activity. IT Curves can provide statistics for server uptime on request.

Integration

MRMS has no dependencies on outside systems. The application can run on most any commercially available personal computer running most browsers. MRMS is extensively tested on Firefox, Chrome, Explorer, and more; IT Curves recommends using Firefox for best results and highest security, but the difference in performance is negligible.

IT Curves uses open APIs to easily integrate with other transportation technologies which clients or their other contractors may use. Common integrations include providing dispatch to third-party fleets, vehicle data collection through OBD-II devices, and the capability to produce custom solutions for other purposes.

Change Management

IT Curves releases quarterly updates to the standard release package. All updates come with a release information package describing the changes and how they may impact the user experience. All updates are optional. All versions of the code are stored by IT Curves in our code management programs to provide ongoing support for all instances of MRMS, allowing our engineers to work on whichever version of the system that each client prefers to run.

Update packages are tested thoroughly in-house and on the IT Curves fleet prior to broad implementation. Results of the testing are included with the release information package. Any custom programming for clients is similarly tested internally prior to implementation.

Data Access

All data produced by each client is the sole property of that client. Access to that data is restricted to only those users with valid credentials as determined by the client. Data for each IT Curves client is stored separately on our cloud and is inaccessible to unauthorized users.

Data can be imported to or exported from the server easily using the data reporting tools included as part of the general release software. Reports can export into multiple formats, and the data is not altered or removed during this process.

IT Curves maintains a comprehensive backup system to ensure that no data is lost even in the event of system failure or physical damage to the servers. Data older than a point as determined by the client can be archived to speed up reporting and system performance. Archived data remains available and accessible through the same reporting methods.

At the end of the contract, all data will be exported for the client in the method and format they prefer. Client proprietary data is completely deleted from the IT Curves cloud after successful transmission of the exported data to the client.

Security

The IT Curves network security team follows the CIA triad of Confidentiality, Integrity, and Availability to ensure up-to-date and active security measures are taken to protect the server and any users from all sorts of internet attacks. Only users with valid login credentials using two-factor authentication may access the program. Clients have complete control not only over who may access their data, but also have a full administrative suite to control which kind of data each of their users may access or change. All access and changes to data are recorded with a time-stamp of the use and the user so that a full history of change and access is maintained.

Any breaches will be reported immediately and full reports, once completed, made available to the client’s IT team or relevant representatives. All information is stored securely in our US-based servers, physically maintained by our network engineering team. Physical access is restricted to only those team members involved in maintaining the servers. Backups are maintained in separate locations in case of natural disaster or other physical damage to the primary servers.

Business Continuity and Disaster Recovery

IT Curves offers its own managed high availability computing cloud with periodic full back up and every hour incremental backup. The IT Curves cloud is secured with full firewall and encryption according to the best practices in data security requirements.

As noted, backups are created both onsite and maintained off-site, allowing for the expeditious restoration of data and services regardless of the data interruption. All servers can execute the MRMS program, so any interruptions to client access to the main server will quickly be resolved by activating another.

Our network engineering team constantly reviews and improves all processes, including but not limited to physical security requirements, backup availability, disaster recovery.

Maintenance Plan

IT Curves offers its own managed high availability computing cloud with periodic full back up and every hour incremental backup. Backups are stored in separate locations in case of disaster, The IT Curves cloud hosting is secured with proper firewall and compliance with modern security requirements. There is no need for on-site maintenance at Grant Transit Authority.

The network operation center continuously monitors all network elements and schedules maintenance as needed. In addition, the network operation team performs quarterly routine maintenance on all servers and other network equipment. These services are carefully planned to not cause any interruption to our customers. Should some interruption be necessary, the network operation team will notify affected customers and arrange for service to occur during off-peak hours.

The release management team and onsite support team make the schedules for releases to add updates to and maintain the software. Releases are conducted in a carefully monitored and supervised process. The team makes sure that there is absolutely no loss of data and the maintenance time is always scheduled for off-peak hours.

IT Curves commits to the highest level of maintenance and support throughout the life of the contract. The costs of maintenance and support are included in the Price Proposal.

Support Plan

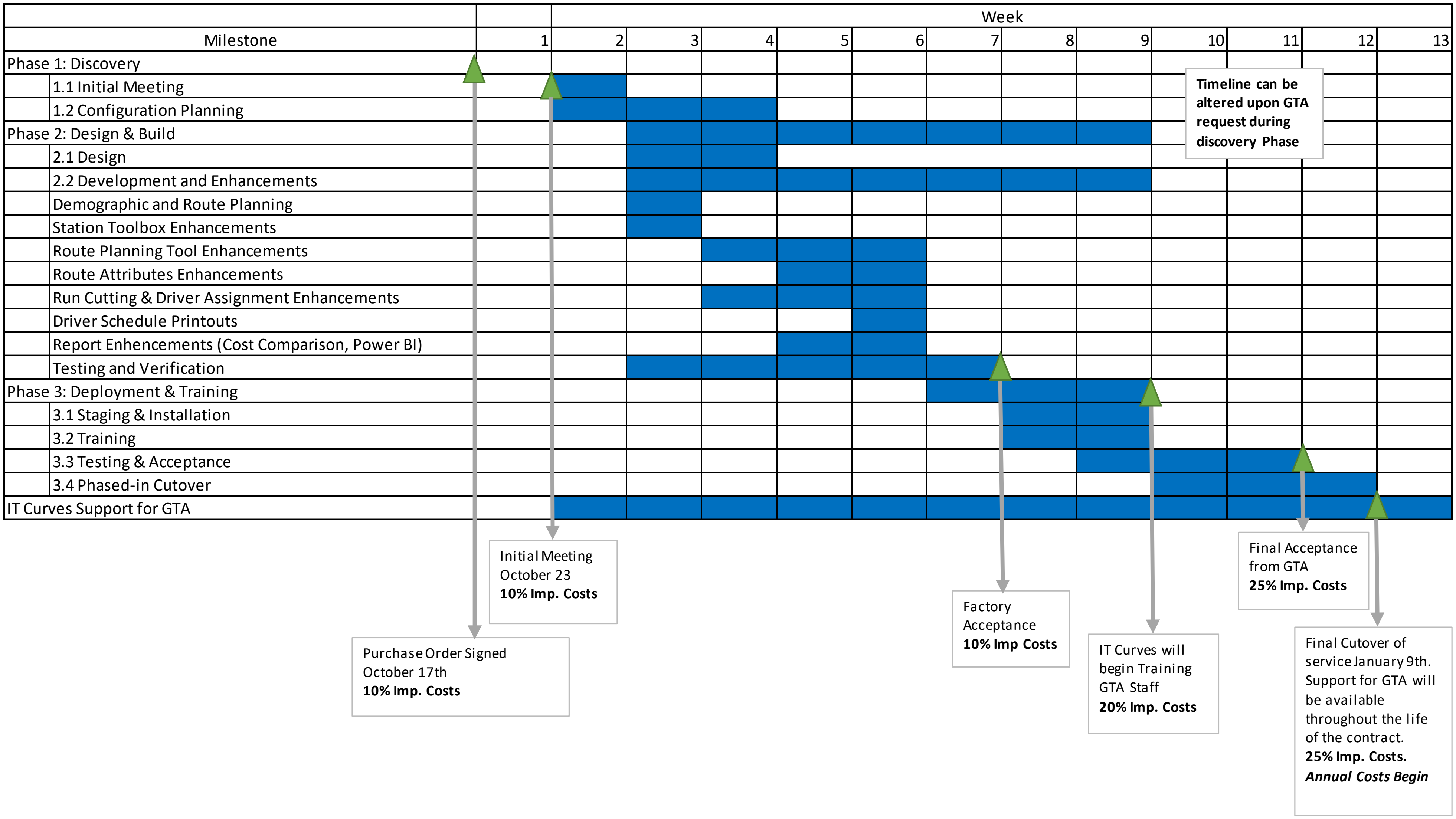
We offer standard technical support from 8AM to 7PM East Coast Time. We provide support both by phone and through the ticketing system built into the software. We typically answer support calls within one minute. We also offer 24-hour emergency support for business-impacting issues according to the following schedule:

Problem Category	Example	Response time	Resolution Time
Critical - Business Impacting	Network down with no work around	1 hour	4 hours
Major - Partial Service Issues	Certain modules or parts of the network are showing issues, but there is a temporary measure that will last until the next day	4	Next day
Minor - Localized Issues	Problem with certain features impacting full capability	4	Next business day
Trivial - Wish List or Enhancements	Minor inconvenience, cosmetic change, or reconfiguration of software features	9:00 AM to 5:00 PM East Coast Time	As agreed with customer for delivery

IT Curves provides phone support for both standard and emergency support.

The best way to work with our customer support is for customers to issue their own tickets on IT Curves' customer ticketing system. This system allows customers to enter their own work tickets, attach documentation, and track the progress of tickets from issue to completion.

5. Project Schedule and Timeliness



a. Warranty, Maintenance

IT Curves offers a comprehensive service and support for all our clients. Our network engineering team maintains the cloud servers on which your instance of MRMS will be hosted and ensures high-availability service and high-level security measures. The IT Curves engineers who support you are the engineers who developed every component offered in this proposal. Your customer support team know every component and not only provide the best maintenance support, but also are fully capable of customizing and enhancing the system as may be needed. IT Curves offers 24/7 emergency support and complete support during business hours.

b. High Compliance to the Scope of Work and Technical Requirements

IT Curves' MRMS system and our ongoing levels of support match and exceed the expectations set out by GTA in the RFP. Our single-system solution to both fixed route and demand response services will be easy to use for staff, allow finalizing any model routes and immediately make it part of the next schedule. New module will not require any additional data import and will be seamlessly integrated into the CAD/AVL system and will be able to transfer data as needed, reduce the burden of multiple logins and systems, and put all maintenance and other requests in the hands of your trusted, personal contacts at IT Curves.

We describe the MRMS system thoroughly, Technical Solution. It also includes our in-detail design showing our direct responses to all questions in RFP, showing our high level of compliance with all GTA's needs.

We understand that GTA would like to move to a new, modern system as of yesterday, and our out-of-the-box software can be implemented quickly and efficiently with minimal involvement from GTA staff. Software-as-a-service, the system is installed remotely in our servers, so GTA will only need to login from anywhere to access MRMS. We will employ a train-the-trainers method to shorten training time and reduce the burden on GTA staff.

Finally, IT Curves complies highly with GTA's preference for an involved and responsive contractor which will provide ongoing support, maintenance, and updates for their software. We offer ongoing support to our clients throughout the life of the contract, providing not only emergency maintenance, but responding to questions, requests, inquiries, and more. Our engineers work with clients to introduce new features and continually improve the system through client feedback, so it will evolve to fit your and GTA's needs year after year.

c. System Installation and Implementation

Our usual schedule for projects of a similar nature tends to be slightly less than what we are presenting to GTA. Given the 12-week time frame for completion, we have opted for an 8-week schedule. This will allow us to patch any bugs prior to January 2025. If GTA would like to alter the proposed timeline, IT Curves is happy to accommodate. Any changes can be discussed during the discovery phase.

- Phase 1 – Initial Meetings: meet on site or virtual, review operation process in detail, collect data files.
- Phase 2 – Configuration and System Setup: create cloud server with staff access rights and data, configured with the input of operations staff.
- Phase 3 – Training and Testing: provide training on all systems to staff who will become the trainers of other staff. Begin testing processes with County staff to validate all functional requirements.
- Phased in Cutover – Go Live: begin transition service to customer based on transition plan.

Currently IT Curves possesses nearly 80% of all the features requested by GTA in this RFP, we expect minor modifications to the software and therefore little development period. As a cloud-based system, minimal work will need to occur on-site and present operations can continue uninterrupted until the new system is ready for use. IT Curves will comply with proposed timeline and deliver all features as per schedule.

The following diagram shows IT Curves' anticipated schedule for the major milestones in system delivery:

IT Curves also offer other installation schedules, including extended schedules which include trial periods for testing on small numbers of vehicles, an immediate cutover to transition all at once, and more pending agency preferences. IT Curves will work with GTA during the Discovery phase to determine the best installation schedule for your needs and preferences.

Our schedule, whether in the twelve-week form or otherwise, includes the following major milestones:

Phase 1 – Discovery

- **Contract Execution:** IT Curves and GTA staff sign the contract agreement, thus commencing this process.
- **Initial Meetings:** To ensure that IT Curves meets all requirements and understands current operation procedures, IT Curves plans to hold meetings with representatives from GTA. We will use these meetings to:
    A. Capture additional detail on project requirements and operational procedures related to programming ERSA for desired functionality, determining the desired standard reports, and more to complete a custom configuration of MRMS.

Phase 2 – Design and Build

- **Update Project Plan Documentation:** Prior to the Kick Off Review, our staff will build the documents necessary for the Kick Off Review. Updates will be presented in the weekly reports and during related meetings.
- **Kick Off Review (KOR):** Within 10 business days of contract signing, IT Curves will present the results of the prior phase for the GTA's review at the KOR. The following documents will be included:
    A. **Statement of Work:** This document will be an enhanced and more detailed version of this Scope of Work. It will include a detailed description of the IT Curves system and display its compliance with mandatory and desirable requirements.
    B. **Project Plan and Schedule:** This document will be an enhanced version of this Project Schedule, including finalized details on dates, hours, and involved personnel pursuant to the reaching the Go Live date.
    C. **Design Specifications:** This document will present the specifications we discovered during the Initial Meetings for GTA approval to begin programming the system according to the Scope of Work.
    D. **Testing Plan:** This document will confirm the testing protocols desired by GTA to affirm the correct and expected functioning of the system by GTA staff, including

the description of the reports which will be produced by IT Curves as part of the testing results.

E. **Deployment Plan:** This document will detail the deployment of the new system along with upgrade of the current system to provide seamless integration between scheduling/runcutting software and current CAD/AVL system.

- **Build and Modify System**: IT Curves engineers will install the system on our cloud according to the procedures described in the Design Specifications document, and Scope of Work. We will also begin importing and converting historical operations data received from the County.
- **Receive Data for Conversion**: As the current deploy system is provided and hosted by IT Curves, there wont be a need for data migration. The scheduling/runcutting software will be integrated with the current CAD/AVL system and can pull the desired data directly through connected APIs.
- **Training Plan and Materials:** IT Curves will prepare a training plan and training materials specific to the City's system and operational procedures.

Phase 3 – Deployment and Training

- **Staging and Installation:** Our staff will fully test and validate the system ourselves during this period so that GTA staff can begin training on and testing the full environment during subsequent phases. We will also begin installing equipment and engage in the second stage of data conversion.
- **Training of County Staff:** A formal training period will be provided according to the Training Plan. IT Curves will train GTA staff in operational concepts, system architecture, and hands-on training with the tools and systems. It will employ a 'train the trainers' approach and include materials so that the trained staff will be able to train new dispatch, administrators, and drivers.
- **Beta Testing:** After providing the customer with access to the IT Curves system cloud and training in how to use the system, a Beta Test period will begin. Beta testing includes test drivers, test customers, and office staff. The test will provide user feedback and ensure that all stakeholders are comfortable with systems operations. The Beta Test period can be extended to accommodate the design, implementation, and further testing of any modifications requested during this phase.
- **System Acceptance Testing:** In parallel with the training and Beta Test, IT Curves engineers and GTA staff will complete a formal Systems Acceptance Test which verifies that all requirements listed in the Statement of Work have been met according to the criteria of the Acceptance Test Procedure Document provided by the City.

- **Phased-in Cutover:** After successful completion of Beta Test and SAT, IT Curves will plan necessary system upgrades to current production system.

Phase 4 – Continuous Support

- **Support:** ongoing support begins immediately after SCO. After final cutover, we will always be reachable through phone, email, and the ticketing system in the IT Curves software.

Weekly Meetings and Reports: Starting immediately after Contract Execution, IT Curves will produce weekly reports for the City's Project Manager, including updates to the Project Schedule as necessary. We will additionally lead meetings as necessary to discuss issue resolution, our adherence to the Project Schedule, and budget.

System Acceptance and Testing

IT Curves builds several rounds of testing into our regular schedule of installation. Our goal, moreover, is to provide our clients with the tools and training necessary to determine the success of the system installation and its configurations. In the first phase, IT Curves will work with GTA to build a Testing Plan to be presented at the KOR, allowing GTA to determine the schedule and testing reports. In the second phase, IT Curves installation staff will work with GTA to determine the protocols and specifications necessary to personalize MRMS for the GTA's needs. This process involves multiple rounds to ensure all specifications are understood. During the third phase, we install the system and train staff on its use prior to the testing period so dispatch and drivers alike know how the system should perform and can judge its installation accordingly. No phase may be considered completed without the active affirmation of GTA and its designated project manager or project management team.

IT Curves is committed to providing the highest level of service throughout the life of the contract. Should any operational defects or other issues which affect system performance arise, our staff is available 24/7 to address the issues.

Training and Documentation

Our trainers have experience with diverse groups and for many needs, and this wide experience has made them particularly effective trainers. IT Curves offers courses both online or in-person. These classes will be available during implementation from our expert trainers and installers for GTA employees.

After the end of the implementation and training schedule, these classes will remain available for your staff to take through online meetings with our training staff. IT Curves will be happy to provide as many classes are necessary, priced according to the costs listed in the Cost Proposal.

Course	Duration (Hours)	Method	Description
CONOP01	4	Class+ Slides Or Video	**System Architecture**: number of modules, descriptions of each module's function.
CONOP02	2	Class+ Slides Or Video	**System Settings:** Customization parameters, system attribute settings. Emphasizes attributes / parameters most applicable to GTA operation.

Copies of all training documentation will be provided to GTA for your future use. Our trainers will remain available for additional sessions at the request of GTA, and our support team can similarly guide staff through solutions to common problems.

Price Proposal and Timeline

Timeline

Payment Percentages are listed on the milestone timeline as x% Imp. Costs (Implementation costs). Refer to Project Schedule and Timeliness.

1 – Purchase Order Signed: 10% Implementation Costs

2 – Initial Meeting/Discovery: 10% Implementation Costs

5 – Factory Acceptance: 10% Implementation Costs

6 – Training begins: 20% Implementation Costs

7 – Final Acceptance from GTA: 25% Implementation Costs

8 – Cutover of Service: 25% Implementation Costs

*Yearly recurring costs begin after Cutover of Service.

6. Fee Schedule

Fee Structure

Cost Table 1: Project Implementation Phase (Capital Budget)

New Feature Add-Ons Integrated into Base Fixed Route System		
Product	**Quantity**	**Cost**
Demographic Route Planning	Add to Site License	$5,250
Enhanced Run Cutting & Driver Assignment Add-On to Base System	Add to Site License	$15,000
Station Toolbox Filter Enhancements	Add to Site License	$1,500
Route Attributes to Routes, Drivers, Vehicles	Add to Site License	$3,750
Route Segment Template through Map and Auto-Update Time/Distance	Add to Site License	$3,750
Adjust Route Path on Map	Add to Site License	$2,250
	Subtotal	**$31,500.00**
Project Delivery Phases		
Product		**Cost**
Design - System Design & Development	20	$ 1,875
Build - Configuration and Integration	20	$ 1,875
Training on New Features and Planner	32	$ 3,000
	Subtotal	**$ 6,750**
	Base System Total	**$ 38,250.00**

Cost Table 2: Annual Recurring Cost (Operating Budget)

Ongoing Annual Cost	
Annual Operating Cost	**Cost**
Software Support, and Maintenance	$ 7,000
Annual Cloud Hosting	$ 4,000
Annual Recurring Total	**$ 11,000**

7. Acknowledgement Form

(This form is intentionally left blank.)

PROPONENT ACKNOWLEDGEMENT

Project Name: TRANSIT ROUTE SCHEDULING/RUNCUTTING SOFTWARE SYSTEM

RFP 24-01

The undersigned agrees to comply with the provisions set forth in this Request for Proposals (including applicable federal requirements) and certifies that all information included in its responsive proposal is true, accurate and complete. Furthermore, the undersigned further agrees, upon execution of an Agreement (if selected), to furnish GTA with services and/or complete the scope of work, in accordance with the terms outlined in this Request for Proposals (including any/all addenda) and in the manner and at the prices proposed.

I declare under penalty of perjury under the laws of the State of Washington that the foregoing is true and correct.

Signed at Gaithersburg, MD 09/06/2024
(City and State) (Date)

Company Name: IT Curves

By: Matthew Mohebbi Title: CEO

8. Non-Collusion Affidavit

(This form is intentionally left blank.)

NON-COLLUSION AFFIDAVIT

Project Name: TRANSIT SCHEDULING AND DISPATCH PLANNING SOFTWARE SYSTEM

RFP 24-01

STATE OF Maryland)
) ss.
COUNTY OF Montgomery)

Matthew Mohebbi, being first duly sworn, on his oath says that the bid above submitted is a genuine and not a sham or a collusive bid, or made in the interest of or on behalf of any person not herein named; and he further states that the said Proponent has not directly or indirectly induced or solicited any other Proponent for the above work or supplies to put in a sham bid, or any other person or corporation to refrain from bidding; and that said Proponent has not in any manner sought by collusion to secure to self advantage over any other Proponent or Proponents.

SIGN HERE ______________________

Subscribed and sworn to before me this 6 day of September, 2022.

GIRMA WOLDE-GIORGIS
NOTARY PUBLIC STATE OF MARYLAND
My Commission Expires March 5, 2023

____

Notary Public in and for the State of
Maryland, residing in Montgomery County

9. FTA Fiscal Year 2024 Certifications & Assurances (Appendix A)

(This form is intentionally left blank.)

FEDERAL FISCAL YEAR 2024 CERTIFICATIONS AND ASSURANCES FOR FTA ASSISTANCE PROGRAMS

(Signature pages alternate to providing Certifications and Assurances in TrAMS.)

Name of Applicant: IT Curves

The Applicant certifies to the applicable provisions of all categories: (*check here*) ✓.

Or,

The Applicant certifies to the applicable provisions of the categories it has selected:

Category		Certification
01	Certifications and Assurances Required of Every Applicant	
02	Public Transportation Agency Safety Plans	
03	Tax Liability and Felony Convictions	
04	Lobbying	
05	Private Sector Protections	
06	Transit Asset Management Plan	
07	Rolling Stock Buy America Reviews and Bus Testing	
08	Urbanized Area Formula Grants Program	
09	Formula Grants for Rural Areas	
10	Fixed Guideway Capital Investment Grants and the Expedited Project Delivery for Capital Investment Grants Pilot Program	
11	Grants for Buses and Bus Facilities and Low or No Emission Vehicle Deployment Grant Programs	

12	Enhanced Mobility of Seniors and Individuals with Disabilities Programs	______
13	State of Good Repair Grants	______
14	Infrastructure Finance Programs	______
15	Alcohol and Controlled Substances Testing	______
16	Rail Safety Training and Oversight	______
17	Demand Responsive Service	______
18	Interest and Financing Costs	______
19	Cybersecurity Certification for Rail Rolling Stock and Operations	______
20	Tribal Transit Programs	______
21	Emergency Relief Program	______

CERTIFICATIONS AND ASSURANCES SIGNATURE PAGE

AFFIRMATION OF APPLICANT

Name of the Applicant: IT Curves

BY SIGNING BELOW, on behalf of the Applicant, I declare that it has duly authorized me to make these Certifications and Assurances and bind its compliance. Thus, it agrees to comply with all federal laws, regulations, and requirements, follow applicable federal guidance, and comply with the Certifications and Assurances as indicated on the foregoing page applicable to each application its Authorized Representative makes to the Federal Transit Administration (FTA) in the federal fiscal year, irrespective of whether the individual that acted on his or her Applicant's behalf continues to represent it.

The Certifications and Assurances the Applicant selects apply to each Award for which it now seeks, or may later seek federal assistance to be awarded by FTA during the federal fiscal year.

The Applicant affirms the truthfulness and accuracy of the Certifications and Assurances it has selected in the statements submitted with this document and any other submission made to FTA, and acknowledges that the Program Fraud Civil Remedies Act of 1986, 31 U.S.C. § 3801 *et seq*., and implementing U.S. DOT regulations, "Program Fraud Civil Remedies," 49 CFR part 31, apply to any certification, assurance or submission made to FTA. The criminal provisions of 18 U.S.C. § 1001 apply to any certification, assurance, or submission made in connection with a federal public transportation program authorized by 49 U.S.C. chapter 53 or any other statute

In signing this document, I declare under penalties of perjury that the foregoing Certifications and Assurances, and any other statements made by me on behalf of the Applicant are true and accurate.

Signature ______________________ Date: 09/06/2024

Name Matthew Mohebbi ______________________ Authorized Representative of Applicant

AFFIRMATION OF APPLICANT'S ATTORNEY

For (Name of Applicant): IT Curves

As the undersigned Attorney for the above-named Applicant, I hereby affirm to the Applicant that it has authority under state, local, or tribal government law, as applicable, to make and comply with the Certifications and Assurances as indicated on the foregoing pages. I further affirm that, in my opinion, the Certifications and Assurances have been legally made and constitute legal and binding obligations on it.

I further affirm that, to the best of my knowledge, there is no legislation or litigation pending or imminent that might adversely affect the validity of these Certifications and Assurances, or of the performance of its FTA assisted Award.

Signature Donna McBride (Sep 6, 2024 15:51 EDT) Date: Sep 6, 2024

Name Donna McBride ______________________ Attorney for Applicant

Each Applicant for federal assistance to be awarded by FTA must provide an Affirmation of Applicant's Attorney pertaining to the Applicant's legal capacity. The Applicant may enter its electronic signature in lieu of the Attorney's signature within TrAMS, provided the Applicant has on file and uploaded to TrAMS this hard-copy Affirmation, signed by the attorney and dated this federal fiscal year.